HANAMANTHAGOUDA POLICEPATIL

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EDUCATION

Indian Institute of Technology Gandhinagar (IITGN) **Jul '23 – May '28**
Dual B.Tech. in Computer Science and Materials Science & Engineering **CPI: 8.0 / 10.0**

RESEARCH EXPERIENCE

Research Intern | Indian Institute of Science (IISc), Bengaluru **May '25 – Jul '25 (Onsite)**

Advisor: Prof. Abhishek Kumar Singh | Materials Research Centre (MRC) *Bengaluru, India*

- **Multimodal Architectures:** Gained familiarity with foundation model approaches including CAST, CLIP, and domain-adaptive pretraining (DAPT) for integrating scientific literature with structured materials data.
- **Instruction Fine-Tuning:** Fine-tuned LLaMA-3 8B using QLoRA and Unsloth on specialized instruction datasets for property screening; optimized context lengths and hyperparameters, increasing classification accuracy from a 52.9% baseline to **90.2%** (metal vs. non-metal). **[GitHub]**
- **Domain-Adaptive Pre-Training (DAPT):** Executed target-domain continual pre-training (CPT) on an extensive scientific corpus using Unsloth; decreased test perplexity to **9.48** and integrated models onto Hugging Face for automated scientific inference. **[GitHub]**
- **Evaluation Pipelines:** Built reproducible automated evaluation benches and tokenization workflows for downstream zero-shot and few-shot materials property prediction.

PUBLICATIONS AND WORKING PAPERS

Adopt, Wait, or Learn? Computational Capital and AI Investment **2025**

Co-authors:Prof. Prateek Sharma (USC) | **Accepted - India Finance Conference' 26 , IIM Bangalore** | **[SSRN]**

- **Dynamic Investment Framework:** Formulated a dynamic model combining real options, internal capital markets, and organizational learning to analyze corporate adoption timing and token-budget allocation across competing business units.
- **Option Value of Waiting:** Demonstrated that computational token consumption generates organizational capital (reusable prompts, automated agents, workflows), creating a key trade-off between delaying adoption and postponing knowledge accumulation.
- **Calibration & Mechanism Design:** Calibrated using enterprise GitHub production data; proved that decentralized allocation mechanisms strictly outperform centralized capital rationing under reporting noise and compute scarcity.

The Crowding Toll on Rented Cognition **2025**

Co-authors: Prof. Prateek Sharma (USC) | **Selected – AFA 2027 Special Project: Collection of AI-Authored Research Papers** | **[Link]**

- **Rented Cognition & Common Shocks:** Examined the delegation of cognitive tasks (screening, pricing, forecasting) to foundation models; proved that while human cognitive errors wash away, rented cognition carries a vendor-level common error component, turning procurement into a common macroeconomic risk factor.
- **Endogenous Factor Structure:** Derived the equilibrium "crowding toll" showing that exposure prices rise with vendor clientele; adopting firms become safer and riskier at once (falling idiosyncratic volatility but surging systematic volatility and higher cost of capital).
- **Concentration-Driven Risk:** Factorized aggregate variance as the product of squared total delegation and vendor concentration (Risk $\propto (\delta_{\text{tot}})^2 \times \text{HHI}$); demonstrated that capability gains raise systematic risk (+**32%** in calibration) primarily by relocating delegation to frontier vendors rather than increasing total adoption.

Real-Time Win Probability Estimation in Twenty20 Cricket: A State-Space Approach **2025**

Co-authors: Prof. Prateek Sharma (USC) & Prof. Amit Arora (IITGN) | *Under Review* – Journal of Quantitative Analysis in Sports | **[SSRN]**

- **State-Space Framework:** Formulated a ball-by-ball continuous-discrete state-space framework tracking three latent match dynamics: Batting Strength (S_t), Scoring Momentum (M_t), and Match Pressure (P_t), updated recursively via **Kalman filtering**.

- **Empirical Calibration:** Evaluated across **1,161 Indian Premier League (IPL) matches** spanning 18 seasons (2008–2025); benchmarked against logistic regression, random forests, and gradient boosting, delivering competitive predictive accuracy with strict temporal coherence.
- **Latent Trajectories:** Proved momentum as the earliest indicator of chase success, pressure as the most persistent discriminator between outcomes, and batting strength as a late-innings confirmation signal.

Agentic Capital Allocation: Data Analysis of On-Chain AI Agent Financial Activity **2025**

Working Paper

- **On-Chain Agent Infrastructure:** Constructed a multi-chain dataset of ERC-8004 registered AI agents on Ethereum and Base, linking agent registrations, operating wallets, ownership transfers, and metadata to reconstruct the full lifecycle of autonomous on-chain agents.
- **Reputation & Financial Reconstruction:** Built agent-day panel datasets covering cumulative ERC-8004 feedback/reputation scores, x402 payment activity (USDC on Base), and wallet-level financial flows, enabling empirical analysis of agent revenue, funding, and counterparty networks.
- **Flow Classification & Sample Construction:** Classified inflows into owner funding, customer/x402 revenue, agent-to-agent payments, and unclassified transfers; validated a random sample of agents and transactions against blockchain records to establish data quality benchmarks.

PROJECTS

Scientific Machine Learning for Physical Systems (Physical PDEs) **2024**

Independent Deep Learning Researcher *PyTorch*

- **Fourier Neural Operators (FNOs) for Turbulence:** Implemented an FNO architecture in PyTorch to simulate 2D Navier-Stokes turbulence. Integrated a targeted noise-injection strategy during autoregressive rollouts, stabilizing long-term trajectories and reducing validation error to **1.04%**. **[GitHub]**
- **Physics-Informed Neural Networks (PINNs):** Formulated PINNs in PyTorch to solve forward field predictions and inverse parameter identification for the 1D Burgers' equation; estimated latent PDE parameters (λ_1, λ_2) from noisy measurements with under **0.22%** error. **[GitHub]**

SKILLS

Programming Languages: Python, C++, C, SQL, Bash

Machine Learning & Libraries: PyTorch, Hugging Face, Scikit-Learn, Unsloth, NumPy, Pandas, SciPy, Matplotlib

Developer Tools & Platforms: Git, Linux/Unix, Docker, Jupyter

LEADERSHIP

Class Representative (MSE '27) | IIT Gandhinagar **2024-25**

Primary academic liaison representing 40+ undergraduate students before departmental faculty; coordinated academic schedules, examinations, and feedback channels.